queries from both red-teaming models, and can deploy a fixed amount of compute corresponding to $n \in \{10000, 20000, \ldots, 90000\}$ Haiku queries. Sonnet is $c \in \{10\text{x}, 20\text{x}, 50\text{x}, 100\text{x}\}$ more expensive than Haiku, so the red-teamer can use either n Haiku queries or n/c Sonnet queries. For example, if $n = 50000$ and $c = 10\text{x}$, the red-teamer must forecast whether 50,000 queries from Haiku will produce a higher elicitation probability than $50000/10\text{x} = 5000$ queries from Sonnet. We use most combinations of m, n and c except for those where $n/c < m$, leaving us with 223 settings.

We evaluate by measuring whether the forecasts correctly identify whether to allocate compute to Sonnet or Haiku. However, in many settings, the worst-query risk over n samples for Haiku is comparable to n/c samples from Sonnet, so the cost of incorrect predictions is low, and may just be due to noise. To account for this, we additionally measure the fraction of correct predictions when the actual difference in worst-query risk is over two orders of magnitude; intuitively, this corresponds to the case where getting the forecast right or wrong is most impactful.

Across all of our settings, we find that our forecast chooses the correct output 63% of the time, compared to 54% for the majority baseline and 50% random chance. However our forecasts help make correct predictions much more frequently when the actual probabilities differ; we achieve an accuracy of 79% when the true difference in the (low) probabilities is more than two orders of magnitude. We include an example where we correctly anticipate that allocating more compute to Haiku is optimal due to the better sampling efficiency, despite the scaling being better for Sonnet in Figure 6.

One challenge in this setting is that our forecasts tend to slightly overestimate the actual probability, and the overestimate grows with the length of the forecast. We think exploring ways to reduce the bias is important subsequent work.

7. Discussion

In this work, we forecast how risks grow with deployment scale by studying the elicitation probabilities of evaluation queries. However, there are many ways to make our forecasts more accurate and practical. For each forecast, we could adaptively test the fit of each extreme-value distribution, model whether our evaluation set captures tail behavior, and add uncertainty estimates to our forecasts. We could also explore making forecasts for a broader range of behaviors on more natural distributions of queries. We think these are exciting areas for subsequent work.

In our experiments, we study deployments that are at most three orders of magnitude larger than evaluation and could in principle be evaluation sets themselves. We do this because simulating ground truth for actual deployment scales is prohibitively expensive; this would require generating millions to billions of queries. However we think our forecasts could seamlessly extrapolate to larger-scale deployments that are intractable to test pre-deployment; for example, curating ground-truth evaluation sets on billions of queries is intractable, but only requires slightly more extrapolation on the log-log plot to make forecasts.

We do not study distribution shifts between evaluation and deployment queries, only shifts in the total number of queries. We hope that for many kinds of risks, historical queries are representative of future ones; model developers can thus construct on-distribution evaluation sets from previous usage, or run small-scale beta tests. However, adversaries in practice may adaptively adjust their queries based on the model, and users may deploy models in new settings based on current capabilities. We think studying how robust our forecasts are to distribution shifts is interesting subsequent work.

Another natural approach to find rare behaviors is to *optimize* for them during evaluation. This could include optimizing over prompts to find a behavior (Jones et al., 2023; Zou et al., 2023), or fine-tuning the model to elicit a behavior (Greenblatt et al., 2024). However, these methods suffer from false positives and false negatives: optimizing can find instances of a behavior that are too rare to ever come up in practice, while optimizers can miss behaviors when they optimize over the wrong attack surface, or do not converge to global optima. Our forecasts also have generalization assumptions to test for rare behaviors; we think trading off between these different generalization assumptions can produce better deployment decisions.

Finally, our method naturally extends to monitoring. The maximum elicitation probability provides a real-time metric for how close models are to producing some undesired behavior, and our scaling laws can be used to forecast how much longer a deployment can continue with low risk. Forecasting in real time also resolves some of the limitations of our setup; we can adaptively test whether we are in the extreme tail, refine our forecasts based on additional evidence, and are less susceptible to distribution shifts. We hope our work better allows developers to preemptively flag deployment risks and make necessary adjustments.

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References

Anthropic. Claude 3 model card, 2024. URL https://assets.anthropic.com/m/61e7d27f8c8f5919/original/Claude-3-Model-Card.pdf. Accessed: 2025-01-24.

Anthropic. Claude 3 haiku release, 2024a. URL https://www.anthropic.com/news/claude-3-haiku.

Anthropic. Claude 3.5 sonnet release, 2024b. URL https://www.anthropic.com/news/claude-3-5-sonnet.

Bengio, Y. et al. International ai safety report 2025. Technical report, AI Safety Institute, January 2025. URL https://assets.publishing.service.gov.uk/media/679a0c48a77d250007d313ee/International_AI_Safety_Report_2025_accessible_f.pdf.

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., and et al., S. A. On the opportunities and risks of foundation models, 2022. URL https://arxiv.org/abs/2108.07258.

Borgeaud, S., Mensch, A., Hoffmann, J., Cai, T., Rutherford, E., Millican, K., Van Den Driessche, G. B., Lespiau, J.-B., Damoc, B., Clark, A., De Las Casas, D., Guy, A., Menick, J., Ring, R., Hennigan, T., Huang, S., Maggiore, L., Jones, C., Cassirer, A., Brock, A., Paganini, M., Irving, G., Vinyals, O., Osindero, S., Simonyan, K., Rae, J., Elsen, E., and Sifre, L. Improving language models by retrieving from trillions of tokens. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S. (eds.), *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 2206–2240. PMLR, 17–23 Jul 2022. URL https://proceedings.mlr.press/v162/borgeaud22a.html.

Brown, B., Juravsky, J., Ehrlich, R., Clark, R., Le, Q. V., Ré, C., and Mirhoseini, A. Large language monkeys: Scaling inference compute with repeated sampling, 2024. URL https://arxiv.org/abs/2407.21787.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33: 1877–1901, 2020.

Brundage, M., Avin, S., Wang, J., Belfield, H., and et al., G. K. Toward trustworthy ai development: Mechanisms for supporting verifiable claims, 2020. URL https://arxiv.org/abs/2004.07213.

Chowdhury, N., Aung, J., Shern, C. J., Jaffe, O., Sherburn, D., Starace, G., Mays, E., Dias, R., Aljubeh, M., Glaese, M., Jimenez, C. E., Yang, J., Liu, K., and Madry, A. Introducing swe-bench verified. *OpenAI*, 2024.

Dubois, Y., Galambosi, B., Liang, P., and Hashimoto, T. B. Length-controlled alpacaeval: A simple way to debias automatic evaluators. *arXiv preprint arXiv:2404.04475*, 2024.

Eastern District of Texas, US District Court. Memorandum and order in case 1:23-cv-00281-mac, 2024. URL https://www.courthousenews.com/wp-content/uploads/2024/11/attorney-sanctioned-for-using-ai-hallucinations.pdf.

Feffer, M., Sinha, A., Deng, W. H., Lipton, Z. C., and Heidari, H. Red-teaming for generative ai: Silver bullet or security theater?, 2024. URL https://arxiv.org/abs/2401.15897.

Flam-Shepherd, D., Zhu, K., and Aspuru-Guzik, A. Language models can learn complex molecular distributions. *Nature Communications*, 13(1):3293, 2022. doi: 10.1038/s41467-022-30839-x. URL https://doi.org/10.1038/s41467-022-30839-x.

Gallegos, I. O., Rossi, R. A., Barrow, J., Tanjim, M. M., Kim, S., Dernoncourt, F., Yu, T., Zhang, R., and Ahmed, N. K. Bias and fairness in large language models: A survey. *Computational Linguistics*, 50(3):1097–1179, September 2024. doi: 10.1162/coli_a_00524. URL https://aclanthology.org/2024.cl-3.8/.

Ganguli, D., Lovitt, L., Kernion, J., Askell, A., Bai, Y., Kadavath, S., Mann, B., Perez, E., Schiefer, N., Ndousse, K., et al. Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned. *arXiv preprint arXiv:2209.07858*, 2022.

Gemini, T., Georgiev, P., Lei, V. I., Burnell, R., Bai, L., Gulati, A., Tanzer, G., Vincent, D., Pan, Z., Wang, S., et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.

Good, I. J. Speculations concerning the first ultraintelligent machine. In *Advances in computers*, volume 6, pp. 31–88. Elsevier, 1966.

Greenblatt, R., Roger, F., Krasheninnikov, D., and Krueger, D. Stress-testing capability elicitation with password-locked models, 2024. URL https://arxiv.org/abs/2405.19550.

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding, 2021. URL https://arxiv.org/abs/2009.03300.

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., Casas, D. d. L., Hendricks, L. A., Welbl, J., Clark, A., et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.

Hughes, J., Price, S., Lynch, A., Schaeffer, R., Barez, F., Koyejo, S., Sleight, H., Jones, E., Perez, E., and Sharma, M. Best-of-n jailbreaking, 2024. URL https://arxiv.org/abs/2412.03556.

Jain, N., Han, K., Gu, A., Li, W.-D., Yan, F., Zhang, T., Wang, S., Solar-Lezama, A., Sen, K., and Stoica, I. Livecodebench: Holistic and contamination free evaluation of large language models for code. *arXiv preprint arXiv:2403.07974*, 2024.

Ji, J., Qiu, T., Chen, B., Zhang, B., Lou, H., Wang, K., Duan, Y., He, Z., Zhou, J., Zhang, Z., Zeng, F., Ng, K. Y., Dai, J., Pan, X., O'Gara, A., Lei, Y., Xu, H., Tse, B., Fu, J., McAleer, S., Yang, Y., Wang, Y., Zhu, S.-C., Guo, Y., and Gao, W. Ai alignment: A comprehensive survey, 2024. URL https://arxiv.org/abs/2310.19852.

Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., and Narasimhan, K. Swe-bench: Can language models resolve real-world github issues?, 2024. URL https://arxiv.org/abs/2310.06770.

Jones, E., Dragan, A., Raghunathan, A., and Steinhardt, J. Automatically auditing large language models via discrete optimization. In Krause, A., Brunskill, E., Cho, K., Engelhardt, B., Sabato, S., and Scarlett, J. (eds.), *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 15307–15329. PMLR, 23–29 Jul 2023.

Kandpal, N., Deng, H., Roberts, A., Wallace, E., and Raffel, C. Large language models struggle to learn long-tail knowledge. In Krause, A., Brunskill, E., Cho, K., Engelhardt, B., Sabato, S., and Scarlett, J. (eds.), *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 15696–15707. PMLR, 23–29 Jul 2023. URL https://proceedings.mlr.press/v202/kandpal23a.html.

Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., and Amodei, D. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Lewkowycz, A., Andreassen, A., Dohan, D., Dyer, E., Michalewski, H., Ramasesh, V., Slone, A., Anil, C., Schlag, I., Gutman-Solo, T., Wu, Y., Neyshabur, B., Gur-Ari, G., and Misra, V. Solving quantitative reasoning problems with language models. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A. (eds.), *Advances in Neural Information Processing Systems*, volume 35, pp. 3843–3857. Curran Associates, Inc., 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/file/18abbeef8cfe9203fdf9053c9c4fe191-Paper-Conference.pdf.

Li, R., Allal, L. B., Zi, Y., Muennighoff, N., and et al., D. K. Starcoder: may the source be with you!, 2023. URL https://arxiv.org/abs/2305.06161.

Li, T., Chiang, W.-L., Frick, E., Dunlap, L., Wu, T., Zhu, B., Gonzalez, J. E., and Stoica, I. From crowdsourced data to high-quality benchmarks: Arena-hard and benchbuilder pipeline. *arXiv preprint arXiv:2406.11939*, 2024.

M. Bran, A., Cox, S., Schilter, O., Baldassari, C., White, A. D., and Schwaller, P. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6 (5):525–535, 2024. doi: 10.1038/s42256-024-00832-8. URL https://doi.org/10.1038/s42256-024-00832-8.

MAA. American invitational mathematics examination - aime. In *American Invitational Mathematics Examination - AIME 2024*, February 2024. URL https://maa.org/maa-invitational-competitions/.

Madani, A., Krause, B., Greene, E. R., Subramanian, S., Mohr, B. P., Holton, J. M., Olmos, J. L., Xiong, C., Sun, Z. Z., Socher, R., Fraser, J. S., and Naik, N. Large language models generate functional protein sequences across diverse families. *Nature Biotechnology*, 41(8): 1099–1106, 2023. doi: 10.1038/s41587-022-01618-2. URL https://doi.org/10.1038/s41587-022-01618-2.

Metta, S., Chang, I., Parker, J., Roman, M. P., and Ehuan, A. F. Generative ai in cybersecurity, 2024. URL https://arxiv.org/abs/2405.01674.

Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., and Gebru, T. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, FAT* '19, pp. 220–229. ACM, January 2019. doi: 10.1145/3287560.3287596. URL http://dx.doi.org/10.1145/3287560.3287596.

National Cyber Security Centre. The near-term impact of ai on the cyber threat, 2025. URL https://www.ncsc.gov.uk/report/impact-of-ai-on-cyber-threat.

Nijkamp, E., Pang, B., Hayashi, H., Tu, L., Wang, H., Zhou, Y., Savarese, S., and Xiong, C. Codegen: An open large language model for code with multi-turn program synthesis, 2023. URL https://arxiv.org/abs/2203.13474.

Omiye, J. A., Lester, J. C., Spichak, S., Rotemberg, V., and Daneshjou, R. Large language models propagate race-based medicine. *npj Digital Medicine*, 6(1):195, 2023. doi: 10.1038/s41746-023-00939-z. URL https://doi.org/10.1038/s41746-023-00939-z.

Omohundro, S. M. The basic ai drives. In *Proceedings of the 2008 Conference on Artificial General Intelligence 2008: Proceedings of the First AGI Conference*, pp. 483–492, NLD, 2008. IOS Press. ISBN 9781586038335.

OpenAI. Openai o1 system card, 2024. URL https://arxiv.org/abs/2412.16720.

OpenAI. Learning to reason with llms, 2024a. URL https://openai.com/index/learning-to-reason-with-llms/. Accessed: 2025-01-29.

OpenAI. Introducing SimpleQA, 2024b. URL https://openai.com/index/introducing-simpleqa/.

Perez, E., Huang, S., Song, F., Cai, T., Ring, R., Aslanides, J., Glaese, A., McAleese, N., and Irving, G. Red teaming language models with language models, 2022. URL https://arxiv.org/abs/2202.03286.

Phan, L., Gatti, A., Han, Z., Li, N., and et al., J. H. Humanity's last exam, 2025. URL https://arxiv.org/abs/2501.14249.

Phuong, M., Aitchison, M., Catt, E., Cogan, S., Kaskasoli, A., Krakovna, V., Lindner, D., Rahtz, M., Assael, Y., Hodkinson, S., Howard, H., Lieberum, T., Kumar, R., Raad, M. A., Webson, A., Ho, L., Lin, S., Farquhar, S., Hutter, M., Deletang, G., Ruoss, A., El-Sayed, S., Brown, S., Dragan, A., Shah, R., Dafoe, A., and Shevlane, T. Evaluating frontier models for dangerous capabilities, 2024. URL https://arxiv.org/abs/2403.13793.

Rein, D., Hou, B. L., Stickland, A. C., Petty, J., Pang, R. Y., Dirani, J., Michael, J., and Bowman, S. R. Gpqa: A graduate-level google-proof q&a benchmark, 2023. URL https://arxiv.org/abs/2311.12022.

Shevlane, T., Farquhar, S., Garfinkel, B., Phuong, M., Whittlestone, J., Leung, J., Kokotajlo, D., Marchal, N., Anderljung, M., Kolt, N., Ho, L., Siddarth, D., Avin, S., Hawkins, W., Kim, B., Gabriel, I., Bolina, V., Clark, J., Bengio, Y., Christiano, P., and Dafoe, A. Model evaluation for extreme risks, 2023. URL https://arxiv.org/abs/2305.15324.

Snell, C., Lee, J., Xu, K., and Kumar, A. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.

Souly, A., Lu, Q., Bowen, D., Trinh, T., Hsieh, E., Pandey, S., Abbeel, P., Svegliato, J., Emmons, S., Watkins, O., et al. A strongreject for empty jailbreaks. *arXiv preprint arXiv:2402.10260*, 2024.

Uesato, J., Kumar, A., Szepesvari, C., Erez, T., Ruderman, A., Anderson, K., Krishmamurthy, Dvijotham, Heess, N., and Kohli, P. Rigorous agent evaluation: An adversarial approach to uncover catastrophic failures, 2018. URL https://arxiv.org/abs/1812.01647.

Wang, Y., Ma, X., Zhang, G., Ni, Y., Chandra, A., Guo, S., Ren, W., Arulraj, A., He, X., Jiang, Z., Li, T., Ku, M., Wang, K., Zhuang, A., Fan, R., Yue, X., and Chen, W. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *CoRR*, abs/2406.01574, 2024. URL https://doi.org/10.48550/arXiv.2406.01574.

Webb, S., Rainforth, T., Teh, Y. W., and Kumar, M. P. A statistical approach to assessing neural network robustness, 2019. URL https://arxiv.org/abs/1811.07209.

Wei, J., Tay, Y., Bommasani, R., Raffel, C., Zoph, B., Borgeaud, S., Yogatama, D., Bosma, M., Zhou, D., Metzler, D., et al. Emergent abilities of large language models. *arXiv preprint arXiv:2206.07682*, 2022.

Weidinger, L., Mellor, J., Rauh, M., Griffin, C., Uesato, J., Huang, P.-S., Cheng, M., Glaese, M., Balle, B., Kasirzadeh, A., Kenton, Z., Brown, S., Hawkins, W., Stepleton, T., Biles, C., Birhane, A., Haas, J., Rimell, L., Hendricks, L. A., Isaac, W., Legassick, S., Irving, G., and Gabriel, I. Ethical and social risks of harm from language models, 2021. URL https://arxiv.org/abs/2112.04359.

Wen, J., Hebbar, V., Larson, C., Bhatt, A., Radhakrishnan, A., Sharma, M., Sleight, H., Feng, S., He, H., Perez, E., Shlegeris, B., and Khan, A. Adaptive deployment of untrusted llms reduces distributed threats, 2024. URL https://arxiv.org/abs/2411.17693.

Wiener, N. Some moral and technical consequences of automation. *Science*, 131(3410):1355–1358, 1960. doi: 10.1126/science.131.3410.1355. URL https://www.science.org/doi/abs/10.1126/science.131.3410.1355.

Wu, G. and Hilton, J. Estimating the probabilities of rare outputs in language models. *arXiv preprint arXiv:2410.13211*, 2024.